

Name: File

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Extension 1

Year 11

Preliminary HSC Assessment Task 2

July 2008

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- Marks may not be awarded for careless or badly arranged work

Question 1	Question 2	Question 3	Question 4	Question 5	TOTAL
/10	/10	/10	/10	/10	/50

Question 1 (10 marks)

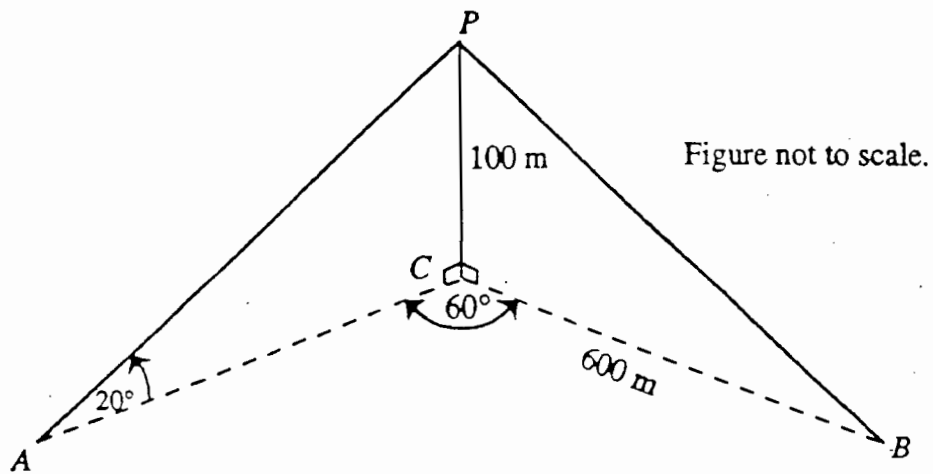
- a) Express $\frac{1}{1-\frac{m}{n}} \div \frac{1}{1-\frac{n}{m}}$ as a single fraction 2
- b) Solve $\frac{1}{x-1} > \frac{1}{6}$ 2
- c) i) Sketch $y = \cos x^\circ$ for $-180 \leq x \leq 180$ 1
- ii) Sketch $y = \frac{1}{2}$ on the same axes as above 1
- iii) Hence, solve $\cos x^\circ > \frac{1}{2}$ for $-180^\circ \leq x \leq 180^\circ$ 2
- d) Given $A(-4, -6)$ and $B(6, -1)$ find the coordinates of $P(x, y)$ such that P divides the interval AB externally in the ratio $3:2$ 2

Question 2 (10 marks) (Start a new page)

- a) i) Write an expansion for $\sin(A + B)$ 1
- ii) By using part i) find the exact value of $\sin 75^\circ$ 2
- b) Differentiate $f(x) = 1 - 2x^2$ by the method of first principles 2

(Question 2 continues over)

c)



Two yachts A and B subtend an angle 60° at the base C of a cliff. From yacht A the angle of elevation of the point P , 100 metres vertically above C , is 20° . Yacht B is 600 metres from C .

- i) Calculate the length AC in exact form. 1
- ii) Calculate the distance between the two yachts. (to nearest metre). 1

- d) The line $y = mx$ makes an angle of 45° with the line $y = 2x - 3$. Find the two possible values for m . 3

Question 3 (10 marks) (Start a new page)

- a) Differentiate the following

i) $y = \frac{2x}{\sqrt{x}}$ 2

ii) $y = (3 - 2x^2)^4$ 2

iii) $y = \frac{x^2}{x+1}$ 2

b) Show that $\frac{d}{dx}(x\sqrt{x+1}) = \frac{3x+2}{2\sqrt{x+1}}$ 2

c) Simplify $\frac{1+\sin x - \cos x}{1+\sin x + \cos x}$ as far as possible in terms of t , where $t = \tan \frac{x}{2}$

Question 4 (10 marks) (Start a new page)

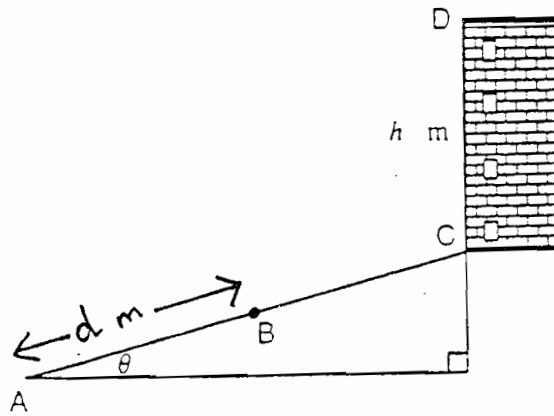
- a) Prove $\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$ 3
- b) Solve
- i) $\sin 2\theta = \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$ 2
- ii) $\cos^2 \theta - \sin^2 \theta = 0.1$ for $0^\circ \leq \theta \leq 180^\circ$
- (θ correct to the nearest minute) 2
- c) i) Write $2\cos x + 3\sin x$ in the form $R\sin(x+\alpha)$ where α is acute and
to the nearest minute, $R > 0$. 1
- ii) Hence, solve $2\cos x + 3\sin x = 1$ for $0^\circ \leq x \leq 360^\circ$ 2

Question 5 (10 marks) (Start a new page)

- a) Given $f(n) = 2^n$ and $g(n) = 3^n$, prove that
- $$\frac{3f(n+1)+2g(n+1)}{6} = f(n) + g(n)$$
- 2
- b) i) Find the equation of the tangent to the semicircle $y = \sqrt{25 - x^2}$ at the
point $P(3, 4)$ 3
- ii) Another tangent to the semicircle above at $Q(0,5)$, meets the tangent at P ,
in the point R . Find the coordinates of R . 1

(Question 5 continues over)

c)



A ramp, ABC rises at an angle of θ degrees. AB is d metres and, at C there is a building of height h metres. The angles of elevation of D from A and B are $(\theta + \alpha)$ and $(\theta + \beta)$ respectively.

i) Find an expression for DB 1

ii) Hence prove that $h = \frac{d \sin \alpha \cdot \sin \beta}{\sin(\beta - \alpha) \cos \theta}$ 1

iii) Deduce that, if $\theta = 15^\circ$, $\alpha = 45^\circ$ and $\beta = 60^\circ$ then $h = d\sqrt{6}$ 2

Question 1

a) $\frac{1}{n-m} \div \frac{1}{\frac{m-n}{m}} = \frac{1}{n-m} \times \frac{m}{m-n} = \frac{m}{(n-m)(m-n)}$

$= \frac{n}{n-m} \times \frac{m-n}{m} = \frac{n}{n-m} \cdot \frac{n}{m} = \frac{n^2}{m(n-m)}$

$= -\frac{n}{n-m} \cdot \frac{n}{m} = -\frac{n^2}{m(n-m)}$

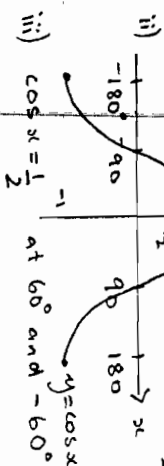
b) $\frac{1}{x-1} > \frac{1}{6}$

$6(x-1) > (x-1)^2$

$6(x-1) - (x-1)^2 > 0$

$(x-1)(7-x) > 0$

$\{x: 1 < x < 7\}$



$\therefore \text{for } \cos x > \frac{1}{2} \quad -60^\circ < x < 60^\circ$

d) A(-4, -6) B(6, -1)

$P\left(\frac{-8+18}{-1}, \frac{-12+3}{-1}\right)$

$P(26, 9)$

Question 2

a) i) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

ii) $\sin 75^\circ = \sin(30^\circ + 45^\circ)$

$= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$

$= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}(\sqrt{3}+1)}{4}$

b) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{1 - 2(x+h)^2 - [1 - 2x^2]}{h}$

$= \lim_{h \rightarrow 0} \frac{1 - 2(x^2 + 2xh + h^2) - 1 + 2x^2}{h}$

$= \lim_{h \rightarrow 0} \frac{-4xh - 2h^2}{h} = \lim_{h \rightarrow 0} (-4x - 2h) = -4x$

c) i) $\tan 30^\circ = \frac{100}{AC}$

ii) $BC = 600 \cdot \frac{1}{\tan 20^\circ}$

$AB = \sqrt{100^2 + 600^2 - 2 \cdot 100 \cdot 600 \cdot \cos 60^\circ}$

$AB = 520 \text{ m}$

d) gradients m and 2

$\tan 45^\circ = \frac{m-2}{1+2m}$

$1 = \frac{m-2}{1+2m} \quad \text{or} \quad -1 = \frac{m-2}{1+2m}$

$1+2m = m-2 \quad -1-2m = m-2$

$m = -3 \quad 1 = 3m$

Question 3

a) i) $y = 2x \Rightarrow y' = 2$

$\therefore \frac{d}{dx}(2x^{1/2}) = x^{-1/2} = \frac{1}{\sqrt{x}}$

ii) $\frac{d}{dx}(3-2x^2)^4 = 4(3-2x^2)^3 \cdot (-4x) = -16x(3-2x^2)^3$

iii) $u = x^2 \Rightarrow u' = 2x$

$\frac{d}{dx}\left(\frac{x^2}{x+1}\right) = \frac{2x(x+1) - x^2}{(x+1)^2} = \frac{x+2x}{(x+1)^2}$

$\frac{d}{dx}\left(\frac{x^2}{x+1}\right) = \frac{x+2x}{(x+1)^2}$

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b) $u = x \Rightarrow v = (x+1)^{1/2}$

$u' = 1 \quad v' = \frac{1}{2}(x+1)^{-1/2} = \frac{1}{2\sqrt{x+1}}$

$\frac{dy}{dx} = \frac{\sqrt{x+1} + \frac{x}{2\sqrt{x+1}}}{2\sqrt{x+1}}$

$= \frac{2(x+1) + x}{2\sqrt{x+1}}$

$\frac{d}{dx}(x\sqrt{x+1}) = \frac{3x+2}{2\sqrt{x+1}}$

c) $\frac{1+t^2}{1-t^2}$

$\frac{1+t^2}{1-t^2} = \frac{1+\sin x - \cos x}{1+\sin x + \cos x}$

$\frac{1+t^2}{1-t^2} = \frac{1+\sin x - \cos x}{1+\sin x + \cos x}$

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$\frac{1+t^2}{1-t^2} = \frac{1+\sin x - \cos x}{1+\sin x + \cos x}$

a) $LHS = \frac{1-\cos 2A}{1+\cos 2A}$

$= \frac{1-(\cos^2 A - \sin^2 A)}{1+\cos^2 A - \sin^2 A}$

$= \frac{1-\cos^2 A + \sin^2 A}{1+\cos^2 A - \sin^2 A}$

$= \frac{(1-\cos^2 A) + \sin^2 A}{(1-\sin^2 A) + \cos^2 A}$

$= \frac{2\sin^2 A}{2\cos^2 A}$

$= \tan^2 A$

$= \tan^2 A$

b) i) $\sin 2\theta = \cos \theta$

$2\sin \theta \cos \theta - \cos \theta = 0$

$\cos \theta (2\sin \theta - 1) = 0$

$\cos \theta = 0 \quad \sin \theta = \frac{1}{2}$

$\therefore \theta = 90^\circ, 270^\circ \quad \theta = 30^\circ$

ii) $\cos^2 \theta - \sin^2 \theta = 0.1$

$\cos 2\theta = 0.1$

$\text{acute } 2\theta = 84^\circ 16'$

$\therefore 2\theta = 84^\circ 16', 275^\circ 44'$

$\theta = 42^\circ 8', 137^\circ 52'$

c) i) $R = \sqrt{4+9} = \sqrt{13}$

$\sqrt{13} \left\{ \frac{2}{\sqrt{13}} \cos x + \frac{3}{\sqrt{13}} \sin x \right\}$

$\sin \alpha = \frac{2}{\sqrt{13}} \quad \cos \alpha = \frac{3}{\sqrt{13}}$

$\therefore \text{Form required}$

$\sqrt{13} \sin(x + 33^\circ 41') = 1$

ii) $(\sqrt{13} \sin(x + 33^\circ 41')) = 1$

$\sin(x + 33^\circ 41') = \frac{1}{\sqrt{13}}$

$x + 33^\circ 41' = 16^\circ 1', 163^\circ 54'$

$x = 130^\circ 13', 342^\circ 25'$

Question 5

a) $f(n) = 2^n$ $g(n) = 3^n$

$$\begin{aligned} \text{LHS} &= \frac{3f(n+1) + 2g(n+1)}{6} \\ &= \frac{3 \cdot 2^{n+1} + 2 \cdot 3^{n+1}}{6} \\ &= \frac{3 \cdot 2^n \cdot 2 + 2 \cdot 3^n \cdot 3}{6} \\ &= \frac{6(2^n + 3^n)}{6} \\ &= 2^n + 3^n \\ &= f(n) + g(n) \\ &= \text{RHS} \end{aligned}$$

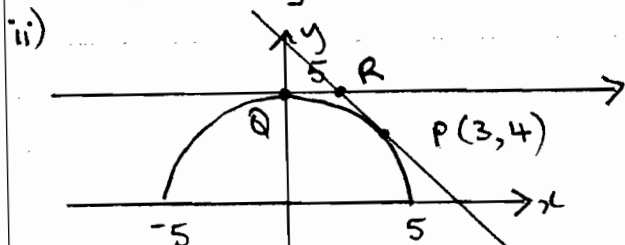
b) i) $y = \sqrt{25 - x^2} = (25 - x^2)^{1/2}$
 $\frac{dy}{dx} = \frac{1}{2} \cdot -2x(25 - x^2)^{-1/2}$
 $= \frac{-x}{\sqrt{25 - x^2}}$

at $P(3, 4)$ $m_T = -\frac{3}{4}$

eqn: $y - 4 = -\frac{3}{4}(x - 3)$

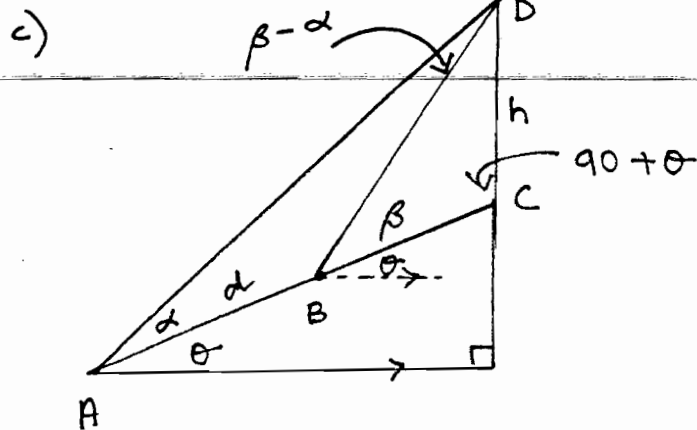
$$4y - 16 = -3x + 9$$

$$3x + 4y - 25 = 0$$



sub $y = 5$ in tang. eqn above

$R\left(\frac{5}{3}, 5\right)$ $3x - 5 = 0$
 $\therefore x = 5/3$



i) In $\triangle ABD$

$$\frac{BD}{\sin \alpha} = \frac{d}{\sin(\beta - \alpha)}$$

$$BD = \frac{d \sin \alpha}{\sin(\beta - \alpha)}$$

ii) In $\triangle BDC$

$$\frac{h}{\sin \beta} = \frac{BD}{\sin(90 + \theta)}$$

$$h = \frac{BD \sin \beta}{\sin(90 + \theta)}$$

note:
 $\sin(90 + \theta) = \cos \theta$

$$\therefore h = \frac{d \sin \alpha \sin \beta}{\sin(\beta - \alpha) \cdot \cos \theta}$$

iii) $h = d \frac{\sin 45^\circ \cdot \sin 60^\circ}{\sin 15^\circ \cdot \cos 15^\circ}$

since
 $\sin 15^\circ = \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ$
 $= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$

$$\cos 15^\circ = \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$$h = d \left[\frac{\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2}}{\frac{\sqrt{3} - 1}{2\sqrt{2}} \cdot \frac{1 + \sqrt{3}}{2\sqrt{2}}} \right]$$

$$h = d \left[\frac{\frac{\sqrt{3}}{2\sqrt{2}} \times \frac{8}{2}}{\frac{\sqrt{3} - 1}{2\sqrt{2}} \cdot \frac{1 + \sqrt{3}}{2\sqrt{2}}} \right]$$